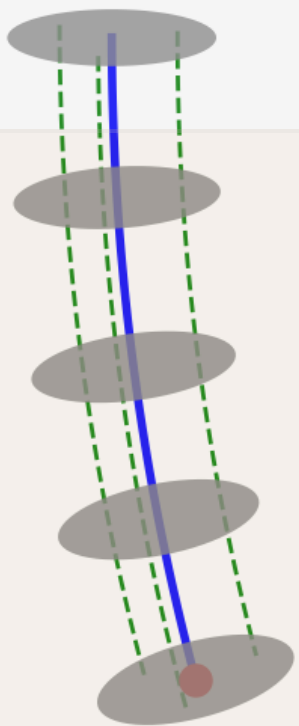


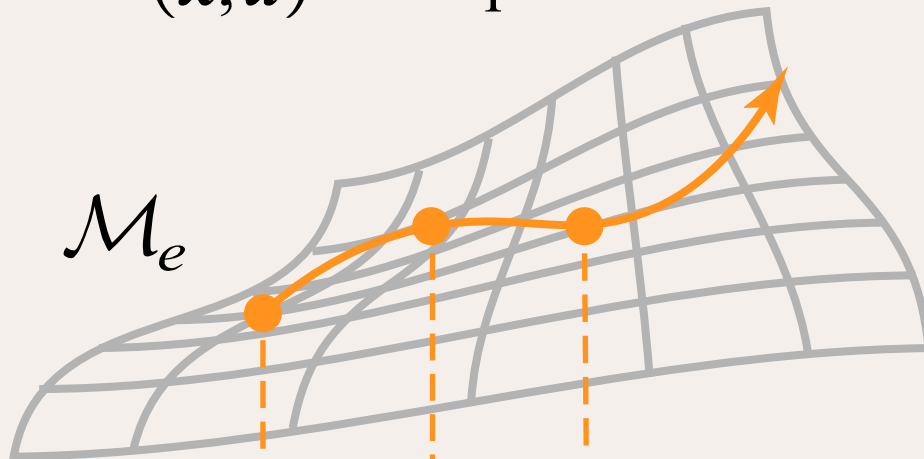
Physical Model



Simulation Model

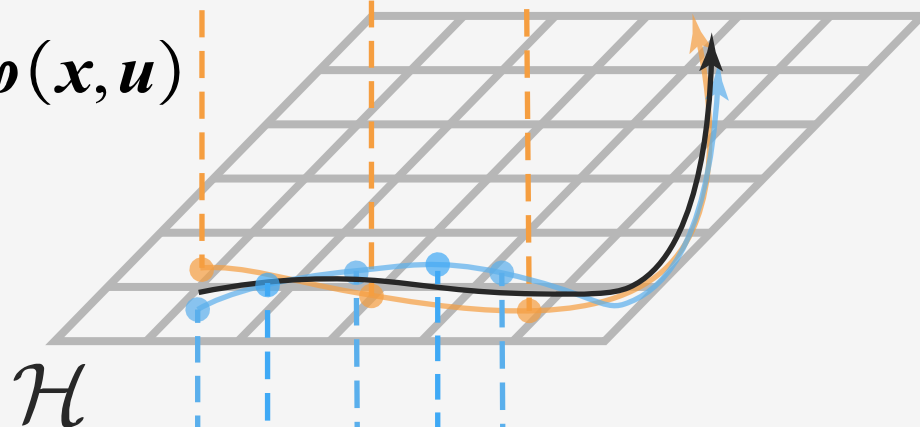


$(\mathbf{x}, \mathbf{u})$ Experimental Data



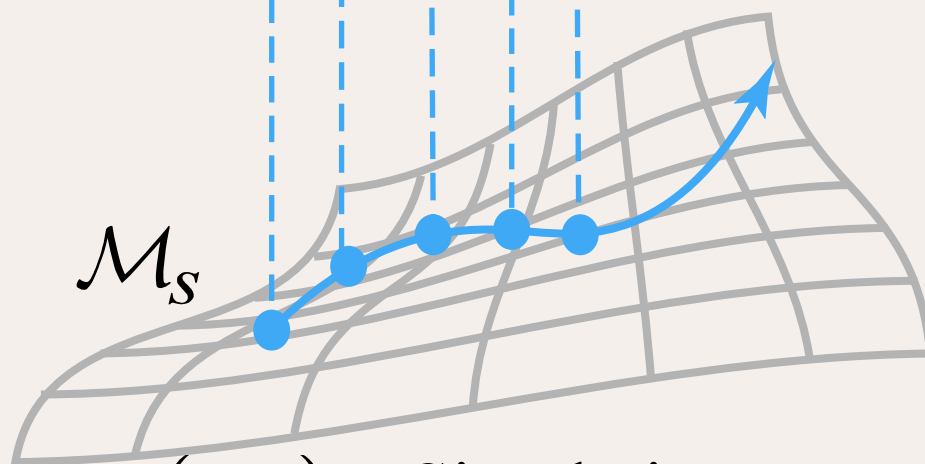
Lifted State

$\varphi(\mathbf{x}, \mathbf{u})$



$\mathcal{M}_s$

$(\mathbf{x}, \mathbf{u})$ Simulation Data



$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) + \mathbf{h}(\mathbf{x}, \mathbf{u})$$

$$\mathcal{K}_{\Delta t}^h$$

$$\varphi(\mathbf{x}_{i+1}, \mathbf{u}_{i+1}) = \mathcal{K}_{\Delta t/2}^f \mathcal{K}_{\Delta t}^h \mathcal{K}_{\Delta t/2}^f \varphi(\mathbf{x}_i, \mathbf{u}_i)$$

$$\mathcal{K}_{\Delta t}^{pi} \approx \mathcal{K}_{\Delta t/2}^f \mathcal{K}_{\Delta t}^h \mathcal{K}_{\Delta t/2}^f$$

$$\varphi(\mathbf{x}_{i+1}, \mathbf{u}_{i+1}) = \mathcal{K}_{\Delta t/2}^f \varphi(\mathbf{x}_i, \mathbf{u}_i)$$

$$\mathcal{K}_{\Delta t/2}^f$$

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$$